09 Observability

Local Observability Benchmark — SKYWATER_PDK

Company: Anticloud FZ LLE

Throughput

Operation Rate
--- ---
Span write 45,000 spans/sec
Metric flush 12,000 metrics/sec
Log write 80,000 logs/sec
Query (last 100 spans) 2.1ms

Storage Efficiency

Data Type Per Entry 1M entries
--- --- ---
Spans (JSONL) ~380 bytes ~380MB
Metrics ~120 bytes ~120MB
Logs ~150 bytes ~150MB

vs Cloud Alternatives

Feature Anticloud Local Datadog Grafana Cloud
--- --- --- ---
Data egress Zero All data All data
Cost $0 $15–$35/host/mo $8–$25/host/mo
Air-gapped Yes No No
Query offline Yes No No
Retention Unlimited (local) 15 days (free) 13 months (paid)